# James Cranley

BM BCh MA MRCP PhD

*Cambridge Stem Cell Institute, Jeffrey Cheah Biomedical Centre, Puddicombe Way, UK, CB2 0AW*

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## Summary


I hold an NIHR-funded Academic Clinical Lecturer post at the University of Cambridge. This affords me 50% time for post-doctoral research (Cambridge Stem Cell Institute) and 50% time to complete Cardiology training (Royal Papworth Hospital). I am due to complete cardiology training in October 2027, sub-specialising in electrophysiology and inherited conditions.

My research integrates electrophysiology with single-cell and spatial genomics to uncover how genetic and cellular heterogeneity drive arrhythmia susceptibility. I aim to establish an independent programme that combines human cardiac multiomics with functional modelling to identify new therapeutic targets for inherited and acquired rhythm disorders.


## Education

**University of Cambridge** — *Cambridge, UK*

PhD in Genomics — *2020 - 2025*

- Funder: Wellcome Trust
- Supervisor: Prof. Sarah Teichmann
- PhD Thesis: *Spatially-resolved multiomic profiling of the human heart in adulthood and development*

**University of Oxford** — *Oxford, UK*

Medical School — *2007 - 2013*

- BM BCh (Distinction)
- BSc (1st class, joint top in year)
- Prizes: Old Member's Scholarship, Matilda Tambyraja (Obstetrics)

**Eton College** — *Windsor, UK*

Secondary School — *2002 - 2007*

- A Levels: Maths, Further Maths, Biology, Chemistry, French, Latin, Greek

## Clinical Experience

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|---|---|
| 2025- | **NIHR Academic Clinical Lectureship in Cardiology**, University of Cambridge |
| 2015-2020 | **NIHR Academic Clinical Fellowship in Cardiology**, East of England |
| 2013-2015 | **Foundation Training**, North-West Thames Deanery |

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## Research Experience

### Post-doctoral (NIHR Academic Clinical Lectureship)

**Stem Cell Institute, University of Cambridge** — *Cambridge, UK*

Supervisor: Prof. Sarah Teichmann — *2025–present*

- Establishing an integrated framework linking single-cell resolved cardiac function with genomic variation, bridging electrophysiology and multiomic datasets to uncover arrhythmia mechanisms.
- Building collaborations across computational and experimental groups to translate cell atlases into mechanistic insight.
- Supervision of Masters and PhD students in the Teichmann Lab. Leading the Heart team subgroup meetings.

### Doctorate (Wellcome Trust PhD in Genomics)

**Wellcome Sanger Institute** *Hinxton, UK*

Supervisor: Prof. Sarah Teichmann *2020–2025*

- Developed single-cell and spatial multiomic maps of the adult and foetal human heart, defining transcriptional and spatial cell states underlying cardiac development and disease.
- Work led to first-author papers in *Nature* (2023) and collaborative outputs in *Nature Cardiovascular Research* and *Science*.

### Pre-doctoral (NIHR Academic Clinical Fellowship)

**Queen Mary University of London** *London, UK*

Supervisor: Prof. Patsy Munroe *2017*

- Performed genome-wide association analyses identifying modulators of the PR interval, contributing to multi-ancestry meta-analyses of cardiac conduction phenotypes.

**Harvard University** *Boston, MA, USA*

Supervisor: Prof. Calum MacRae *2016*

- Investigated non-canonical genomic and phenotypic predictors of coronary artery disease within translational cardiovascular genomics programmes.

**University of Cambridge** *Cambridge, UK*

Supervisor: Prof. Chris Huang *2015*

- Applied whole-cell patch clamping to study the impact of $Pgc1^{\beta}$ deficiency on cardiac sodium current kinetics, linking bioenergetic dysfunction with arrhythmogenic substrates.

### Undergraduate (Intercalated BSc, Physiology)

**University of Oxford** *Oxford, UK*

Supervisors: Prof. Neil Herring, Prof. David Paterson *2011*

- Explored the role of galanin in cardiac autonomic regulation, identifying neuromodulatory effects on vagal tone (published in *J Mol Cell Cardiol*, 2012).

## Selected Publications

H-index 13 | Total citations 1153 | ORCID 0000-0002-0408-5801

Selected publications illustrating contributions to single-cell and spatial cardiac genomics, developmental biology, and clinical cardiology. Full publication list available via Google Scholar

### Co-First Author

**Cranley J***, Kanemaru K*, Bayraktar S*, Knight-Schrijver V, Hulbert R, Lana-Elola E, Aoidi R, Pett JP, Wilbrey-Clark A, Polanski K, Dabrowska M, Mulas I, Johnson H, Richardson L, Semprich CI, Kapuge R, Perera S, He X, Ho SY, Yayon N, Tuck L, Roberts K, Palmer JA, Davaapil H, Gambardella L, Patel M, Tyser R, Bernardo AS, Tybulewicz VLJ, Sinha S, Teichmann SA. 2024. Developmental Dynamics of Human Cardiogenesis: A multi-omic reference and its disruption in Trisomy 21. DOI Under review at *Nature*.

Bayraktar S*, **Cranley J***, Kanemaru K*, Knight-Schrijver V, Colzani M, Davaapil H, Lee JCM, Polanski K, Richardson L, Semprich CI, Kapuge R, Dabrowska M, Mulas I, Perera S, Patel M, Ho SY, He X, Tyser R, Gambardella L, Teichmann SA, Sinha S. 2024. High-resolution atlas of the developing human heart and the great vessels. DOI Under review at *Nature Cardiovascular Research*.

Kanemaru K*, **Cranley J***, Muraro D, Miranda AMA, Ho SY, Wilbrey-Clark A, Patrick Pett J, Polanski K, Richardson L, Litvinukova M, Kumasaka N, Qin Y, Jablonska Z, Semprich CI, Mach L, Dabrowska M, Richoz N, Bolt L, Mamanova L, Kapuge R, Barnett SN, Perera S, Talavera-López C, Mulas I, Mahbubani KT, Tuck L, Wang L, Huang MM, Prete M, Pritchard S, Dark J, Saeb-Parsy K, Patel M, Clatworthy MR, Hübner N, Chowdhury RA, Noseda M, Teichmann SA. 2023. Spatially resolved multiomics of human cardiac niches. *Nature*, 619(7971): 801–810. DOI

### Collaborative

Troulé K*, Petryszak R, Cakir B, **Cranley J**, Harasty A, Prete M, Tuong ZK, Teichmann SA, Garcia-Alonso L, Vento-Tormo R. 2025. CellPhoneDB v5: inferring cell–cell communication from single-cell multiomics data. *Nature Protocols*. DOI

Nicastro M*, Vermeer AM, Postema PG, Tadros R, Bowling FZ, Aegisdottir HM, Tragante V, Mach L, Postma AV, Lodder EM, van Duijvenboden K, Zwart R, Beekman L, Wu L, Jurgens SJ, van der Zwaag PA, Alders M, Allouba M, Aguib Y, Santome

JL, de Una D, Monserrat L, Miranda AM, Kanemaru K, **Cranley J**, van Zeggeren IE, Aronica EM, Ripolone M, Zanotti S, Sveinbjornsson G, Ivarsdottir EV, Hólm H, Guðbjartsson DF, Skúladóttir ÁT, Stefánsson K, Nadauld L, Knowlton KU, Ostrowski SR, Sørensen E, Vesterager Pedersen OB, Ghouse J, Rand SA, Bundgaard H, Ullum H, Erikstrup C, Aagaard B, Bruun MT, Christiansen M, Jensen HK, Carere DA, Cummings CT, Fishler K, Tørring PM, Brusgaard K, Juul TM, Saaby L, Winkel BG, Mogensen J, Fortunato F, Comi GP, Ronchi D, van Tintelen JP, Noseda M, Airola MV, Christiaans I, Wilde AA, Wilders R, Clur SA, Verkerk AO, Bezzina CR, Lahrouchi N. 2025. Bi-allelic variants in POPDC2 cause an autosomal recessive syndrome presenting with cardiac conduction defects and hypertrophic cardiomyopathy. *The American Journal of Human Genetics*, 112(7): 1681–1698. DOI

To K*, Fei L*, Pett JP, Roberts K, Blain R, Polański K, Li T, Yayon N, He P, Xu C, **Cranley J**, Moy M, Li R, Kanemaru K, Huang N, Megas S, Richardson L, Kapuge R, Perera S, Tuck E, Wilbrey-Clark A, Mulas I, Memi F, Cakir B, Predeus AV, Horsfall D, Murray S, Prete M, Mazin P, He X, Meyer KB, Haniffa M, Barker RA, Bayraktar O, Chédotal A, Buckley CD, Teichmann SA. 2024. A multi-omic atlas of human embryonic skeletal development. *Nature*, 635(8039): 657–667. DOI

Barnett SN*, Cujba AM*, Yang L*, Maceiras AR, Li S, Kedlian VR, Pett JP, Polanski K, Miranda AMA, Xu C, **Cranley J**, Kanemaru K, Lee M, Mach L, Perera S, Tudor C, Joseph PD, Pritchard S, Toscano-Rivalta R, Tuong ZK, Bolt L, Petryszak R, Prete M, Cakir B, Huseynov A, Sarropoulos I, Chowdhury RA, Elmentaite R, Madissoon E, Oliver AJ, Campos L, Brazovskaja A, Gomes T, Treutlein B, Kim CN, Nowakowski TJ, Meyer KB, Randi AM, Noseda M, Teichmann SA. 2024. An organotypic atlas of human vascular cells. *Nature Medicine*, 30(12): 3468–3481. DOI

Knight-Schrijver VR*, Davaapil H, Bayraktar S, Ross ADB, Kanemaru K, **Cranley J**, Dabrowska M, Patel M, Polanski K, He X, Vallier L, Teichmann S, Gambardella L, Sinha S. 2022. A single-cell comparison of adult and fetal human epicardium defines the age-associated changes in epicardial activity. *Nature Cardiovascular Research*. DOI

Reichart D*, Lindberg EL*, Maatz H, Miranda AMA, Viveiros A, Shvetsov N, Gärtner A, Nadelmann ER, Lee M, Kanemaru K, Ruiz-Orera J, Strohmenger V, DeLaughter DM, Patone G, Zhang H, Woehler A, Lippert C, Kim Y, Adami E, Gorham JM, Barnett SN, Brown K, Buchan RJ, Chowdhury RA, Constantinou C, **Cranley J**, Felkin LE, Fox H, Ghauri A, Gummert J, Kanda M, Li R, Mach L, McDonough B, Samari S, Shahriaran F, Yapp C, Stanasiuk C, Theotokis PI, Theis FJ, Bogaerdt AVD, Wakimoto H, Ware JS, Worth CL, Barton PJR, Lee YA, Teichmann SA, Milting H, Noseda M, Oudit GY, Heinig M, Seidman JG, Hubner N, Seidman CE. 2022. Pathogenic variants damage cell composition and single cell transcription in cardiomyopathies. *Science*, 377(6606). DOI

Miranda AMA*, Janbandhu V, Maatz H, Kanemaru K, **Cranley J**, Teichmann SA, Hübner N, Schneider MD, Harvey RP, Noseda M. 2022. Single-cell transcriptomics for the assessment of cardiac disease. *Nature Reviews Cardiology*. DOI

Ntalla I*, Weng LC, Cartwright JH, Hall AW, Sveinbjornsson G, Tucker NR, Choi SH, Chaffin MD, Roselli C, Barnes MR, Mifsud B, Warren HR, Hayward C, Marten J, **Cranley J**, Concas MP, Gasparini P, Boutin T, Kolcic I, Polasek O, Rudan I, Araujo NM, Lima-Costa MF, Ribeiro ALP, Souza RP, Tarazona-Santos E, Giedraitis V, Ingelsson E, Mahajan A, Morris AP, Del Greco M F, Foco L, Gögele M, Hicks AA, Cook JP, Lind L, Lindgren CM, Sundström J, Nelson CP, Riaz MB, Samani NJ, Sinagra G, Ulivi S, Kähönen M, Mishra PP, Mononen N, Nikus K, Caulfield MJ, Dominiczak A, Padmanabhan S, Montasser ME, O'Connell JR, Ryan K, Shuldiner AR, Aeschbacher S, Conen D, Risch L, Thériault S, Hutri-Kähönen N, Lehtimäki T, Lyytikäinen LP, Raitakari OT, Barnes CLK, Campbell H, Joshi PK, Wilson JF, Isaacs A, Kors JA, van Duijn CM, Huang PL, Gudnason V, Harris TB, Launer LJ, Smith AV, Bottinger EP, Loos RJF, Nadkarni GN, Preuss MH, Correa A, Mei H, Wilson J, Meitinger T, Müller-Nurasyid M, Peters A, Waldenberger M, Mangino M, Spector TD, Rienstra M, van de Vegte YJ, van der Harst P, Verweij N, Kääb S, Schramm K, Sinner MF, Strauch K, Cutler MJ, Fatkin D, London B, Olesen M, Roden DM, Benjamin Shoemaker M, Gustav Smith J, Biggs ML, Bis JC, Brody JA, Psaty BM, Rice K, Sotoodehnia N, De Grandi A, Fuchsberger C, Pattaro C, Pramstaller PP, Ford I, Wouter Jukema J, Macfarlane PW, Trompet S, Dörr M, Felix SB, Völker U, Weiss S, Havulinna AS, Jula A, Sääksjärvi K, Salomaa V, Guo X, Heckbert SR, Lin HJ, Rotter JI, Taylor KD, Yao J, de Mutsert R, Maan AC, Mook-Kanamori DO, Noordam R, Cucca F, Ding J, Lakatta EG, Qian Y, Tarasov KV, Levy D, Lin H, Newton-Cheh CH, Lunetta KL, Murray AD, Porteous DJ, Smith BH, Stricker BH, Uitterlinden A, van den Berg ME, Haessler J, Jackson RD, Kooperberg C, Peters U, Reiner AP, Whitsel EA, Alonso A, Arking DE, Boerwinkle E, Ehret GB, Soliman EZ, Avery CL, Gogarten SM, Kerr KF, Laurie CC, Seyerle AA, Stilp A, Assa S, Abdullah Said M, Yldau van der Ende M, Lambiase PD, Orini M, Ramirez J, Van Duijvenboden S, Arnar DO, Gudbjartsson DF, Holm H, Sulem P, Thorleifsson G, Thorolfsdottir RB, Thorsteinsdottir U, Benjamin EJ, Tinker A, Stefansson K, Ellinor PT, Jamshidi Y, Lubitz SA, Munroe PB. 2020. Multi-ancestry GWAS of the electrocardiographic PR interval identifies 202 loci underlying cardiac conduction. *Nature Communications*, 11(1). DOI

### Earlier Work

**Cranley J***, MacRae CA. 2018. A New Approach to an Old Problem: One Brave Idea. *Circulation Research*, 122(9): 1172–1175. DOI

Herring N*, **Cranley J**, Lokale MN, Li D, Shanks J, Alston EN, Girard BM, Carter E, Parsons RL, Habecker BA, Paterson DJ. 2012. The cardiac sympathetic co-transmitter galanin reduces acetylcholine release and vagal bradycardia: Implications for neural control of cardiac excitability. *Journal of Molecular and Cellular Cardiology*, 52(3): 667–676. DOI

## Awards, Fellowships, & Grants

### Fellowships

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| 2025–present | **NIHR Academic Clinical Lectureship in Cardiology**, National Institute for Health and Care Research (NIHR) |
| 2015–2020 | **NIHR Academic Clinical Fellowship in Cardiology**, National Institute for Health and Care Research (NIHR) |
| 2020–2025 | **Wellcome Trust PhD for Clinicians Fellowship**, Wellcome Trust |

### Prizes and Awards

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| 2012 | **Old Members' Academic Scholarship**, University of Oxford |
| 2012 | **Matilda Tambyraja Prize (Obstetrics)**, University of Oxford |
| 2013 | **Distinction in Final Medical Examinations**, University of Oxford |
| 2010 | **First Class Honours in Intercalated BSc (joint top in year)**, University of Oxford |
| 2003 | **Academic Oppidan Scholarship**, Eton College |

### Grants

Contributed to the preparation and scientific development of several large-scale collaborative grant applications within the Teichmann Lab, and currently applying for independent early-career funding.

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| 2024–present | **Special Project Award:** ***Immune Processes in Atherosclerosis*** **(contributor)**, British Heart Foundation | *£300,000* |
| 2024–present | **Strategic Longer and Larger Grant:** ***CellTalkHHD*** **(contributor)**, Biotechnology and Biological Sciences Research Council (BBSRC) | *£2,000,000* |
| 2024–present | **BioImaging Technology Development Award:** ***OCMI*** **(contributor)**, Wellcome Trust | *£1,000,000* |

## Qualifications

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| 2025 | **Clinical Research, Education & Leadership module**, University of Cambridge |
| 2019 | **Post-graduate Certificate in Medical Education (PGCME)**, University of Cambridge |
| 2018 | **Adult Life Support (ALS) Instructor**, Resuscitation Council (UK) |
| 2015 | **Membership (MRCP)**, Royal College of Physicians (London) |

## Presentations

*Risk stratification in Brugada Syndrome: a case presentation*. **British Cardiac Society**. 2025. Manchester, UK.

*The Devil is in the Detail: nodofascicular VT*. **British Heart Rhythm Society**. 2025. Edinburgh, UK.

*What you're missing matters: integrating short and long-read scRNA-seq*. **Oxford Nanopore Symposium**. 2024. Oxford, UK.

*Spatial Transcriptomics in Human Cell Atlasing*. **SpatialBiology 2023**. 2023. Online.

*Gene regulatory networks underlying cardiogenesis*. **European Society of Cardiology Congress**. 2023. Amsterdam, NL.

*A spatially-resolved multiomic atlas of human cardiac development*. **HCA General Meeting**. 2023. Toronto, CA.

*The adult and developing heart, one cell at a time*. **The Royal Society**. 2023. London, UK.

*The Hopes and Fears Lab (Public Engagement event)*. **Cambridge Alumni Festival**. 2022. Cambridge, UK.

*Single-cell atlasing of the heart*. **BHF Symposium**. 2022. Online.

*A novel variant causing JLNS, but not LQT1*. **Heart Rhythm Congress**. 2019. Birmingham, UK.

*Effusive-constrictive pericarditis – a complication of PCI*. **British Society of Cardiac Imaging**. 2019. Cambridge, UK.

*Haematoma risk post-angiography*. **Society for Cardiac Interventions & Angiography (SCAI)**. 2014. San Diego, USA.

## Teaching and Supervision

*Adult Life Support (ALS) teaching*. **Resuscitation Council UK**. 2025. Nationwide, UK.

*Core Cardiology course for general medics (Course Director)*. **Royal Papworth Hospital**. 2025. Cambridge, UK.

*Cardiology Webinars for trainees (BJCA.tv series)*. **East of England Deanery**. 2020. Online.

*Undergraduate Physiology supervisions (HOM)*. **Gonville & Caius College, University of Cambridge**. 2017. Cambridge, UK.

*Medical Student Teaching Programme (teaching schedule for rotating students)*. **Harefield Hospital**. 2015. London, UK.

## Management, Leadership & Responsibilities

**Medical Director's Office Fellowship**. Completed a 12-month fellowship in the Medical Director's Office at Royal Papworth Hospital, gaining insight into clinical governance, leadership, and quality improvement. Led a project to improve equity in registrar training opportunities.

**Cardiology rota co-ordinator**. In this 12 month post I was responsible for constructing the termly and weekly rotas and liaison with the consultant body. During this role I gained experience of personnel management, navigated 2 strikes, filled a large rota gap, and successful proposed a business case and created a job advert for a clinical fellow.

**BJCA representative to the BHRS**. I represented the BJCA interest (electrophysiology and devices trainees) at BHRS committee meetings. I also attended BJCA committee meetings.

**Trainee representative for East of England (Digital & Training Days**. I constructed a website for cardiology trainees as well as organising regional training events.